Fast Computation of Multivariate Subresultants

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Outline

Introduction

Problem

Given $f, g \in R[x]$ with $\deg(f) = p > q = \deg(g)$, compute

$$\gcd(f, g), \qquad \mathrm{Res}(f, g), \qquad \left(\mathrm{sResP}_i(f, g)\right)_{0 \le i \le q}.$$

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Background

Univariate and multivariate polynomials, GCD, resultants, subresultants and subresultant polynomials.

Euclidean Algorithm: Correct but Costly

Euclidean GCD:

$$r_0 = f, \qquad r_1 = g, \qquad r_{i+1} = r_{i-1} \bmod r_i.$$

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$$g = 3x^6 + 5x^4 - 4x^2 - 9x + 21.$$

$$r_2 = -\frac{5}{9}x^4 + \frac{1}{9}x^2 - \frac{1}{3}, \qquad r_3 = -\frac{117}{25}x^2 - 9x + \frac{441}{25},$$
$$r_4 = \frac{233150}{19773}x - \frac{102500}{6591}, \qquad r_5 = -\frac{1288744821}{543589225}.$$

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Key property

The subresultant subsequence of polynomials is proportional to the Euclidean remainder sequence, and GCD is the final non-zero subresultant polynomial.

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For the same f and g:

$$\mathrm{sResP}_5(f, g) = 15x^4 - 3x^2 + 9,$$
$$\mathrm{sResP}_3(f, g) = 65x^2 + 125x - 245,$$
$$\mathrm{sResP}_2(f, g) = 9326x - 12300,$$
$$\mathrm{sResP}_0(f, g) = 260708.$$

$$\gcd(f, g) = 1.$$

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Observed performance

Speedups up to $7\times$ on benchmarks with total degree of bitsize 6 and coefficient bit size 50, both on FMPZ and FMPQ.

Subresultant Polynomials

For $0 \leq i \leq q$ and $0 \leq j \leq i$, let $M_{i,j}(f, g)$ be the corresponding Sylvester-Habicht minor.

Definition

$$\mathrm{sResP}_i(f, g) = \sum_{j=0}^{i} \det(M_{i,j}(f, g))x^j.$$

Non-defective case

$$\deg\big(\mathrm{sResP}_i(f, g)\big) = i.$$

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- Non-zero subresultants encode Euclidean remainders.
- Defective subresultants reveal degree drops.
- Zero subresultants indicate gaps in the sequence.
- Last non-zero subresultant polynomial equals the GCD.

Consequence

The GCD and resultant can be computed without division in the fraction field.

Multivariate Setting

Let

$$f, g \in \mathbb{Z}[x_1, \ldots, x_n].$$

View them as univariate polynomials in x_1:

$$f, g \in R[x_1], \qquad R = \mathbb{Z}[x_2, \ldots, x_n].$$

Goal

Compute

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Symbolic computation over

$$R = \mathbb{Z}[x_2, \ldots, x_n]$$

can be expensive.

Instead, choose evaluation points

$$\eta \in \mathbb{Z}^{n-1}$$

and compute in

$$\mathbb{Z}[x_1].$$

Backbone of the Approach

Specialization of the Resultant
Let

$$h = \mathrm{lc}_{x_1}(f)\,\mathrm{lc}_{x_1}(g).$$

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Degree Bound
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Bound

$$\deg\big(\mathrm{Res}_{x_1}(f, g)\big) \leq D.$$

Interpolation cost

For $n = 2$, at most $D + 1$ evaluation points are needed.

Proposed Bivariate Resultant Algorithm

For $f, g \in \mathbb{Z}[x_1, x_2]$:

$N = \deg_{x_1}(f)\deg_{x_2}(g) + \deg_{x_1}(g)\deg_{x_2}(f)$.

1. Choose $N+1$ good evaluation points.
2. Compute

$$y_i = \mathrm{Res}(f(x_i), g(x_i)).$$

3. Interpolate

$$R = \mathrm{interpolate}(x_i, y_i, N+1).$$

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Implemented strategies:

- Random Points
- Random Positive Points
- Small Random Points
- Small Random Positive Points
- Ordered Points
- Ordered Positive Points

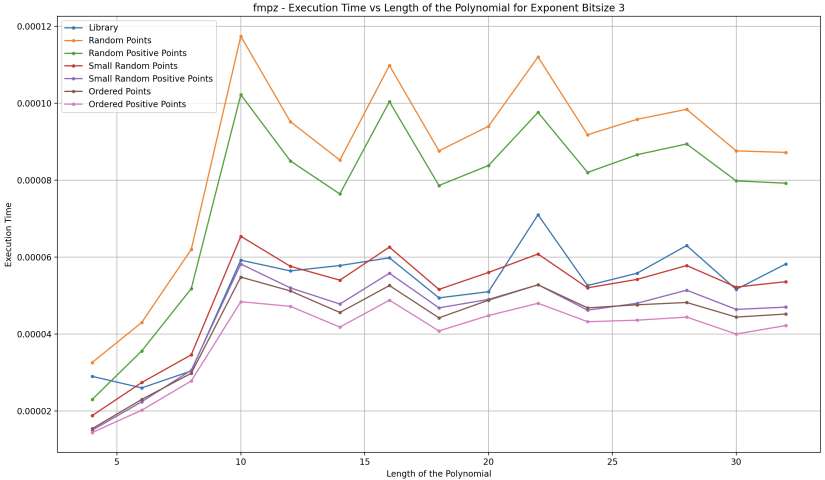
Small and ordered points usually reduce interpolation cost.

FMPZ Benchmarks

Benchmark Setup

- Apple MacBook, 8-core Apple M2 SoC
- 16 GB unified memory
- Average of 5 independent trials
- Coefficient bit size fixed at 50
- Comparison against FLINT bivariate resultant computation

FMPZ Benchmarks



Bitsize 3

Random interpolation points with large bitsize are much slower.

Not too much difference between other benchmarks.

Length of the polynomial does not affect the performance significantly.

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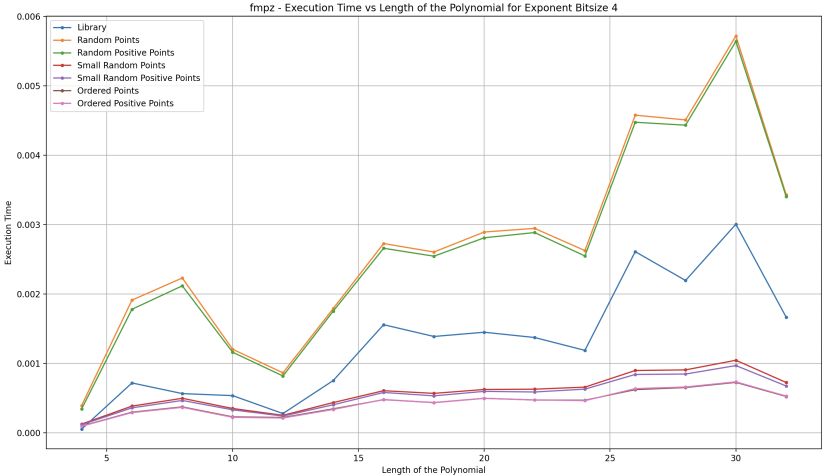

Bitsize 4

Library implementation is slower than our algorithm.

Not too much difference between random interpolation with small bitsize and ordered points.

Execution time increases as the length of the polynomial increases.

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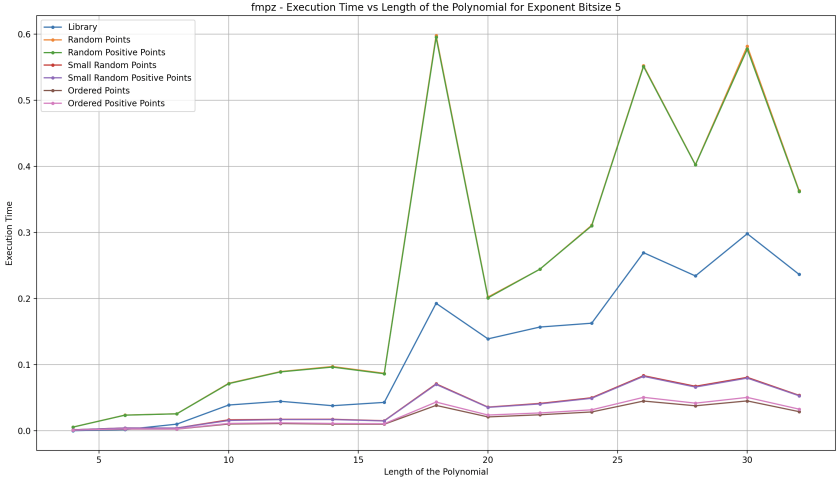


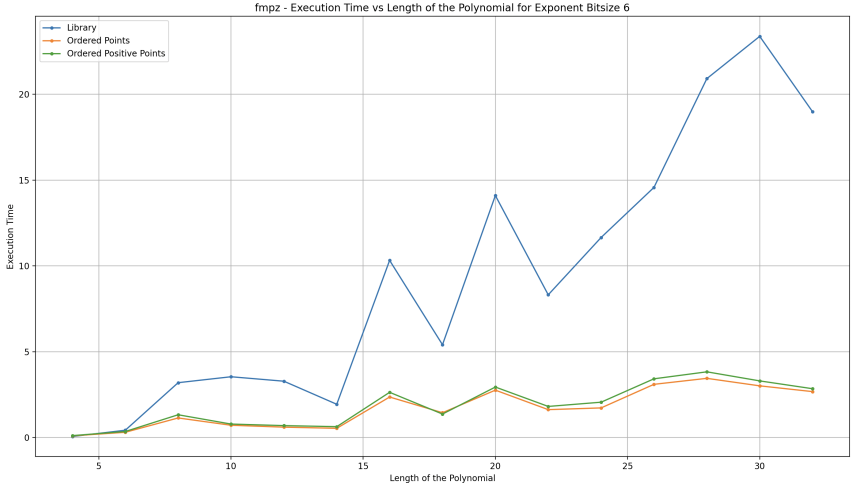
Bitsize 5

Point bitsize strongly affects interpolation time.

We observe high fluctuations in the performance because of random polynomial generation.

Not an accurate measurement of dependance to the length, but the ratio between different methods.

FMPZ Benchmarks



Bitsize 6

We remove random interpolation methods.

We observe 6x performance difference between our methods.

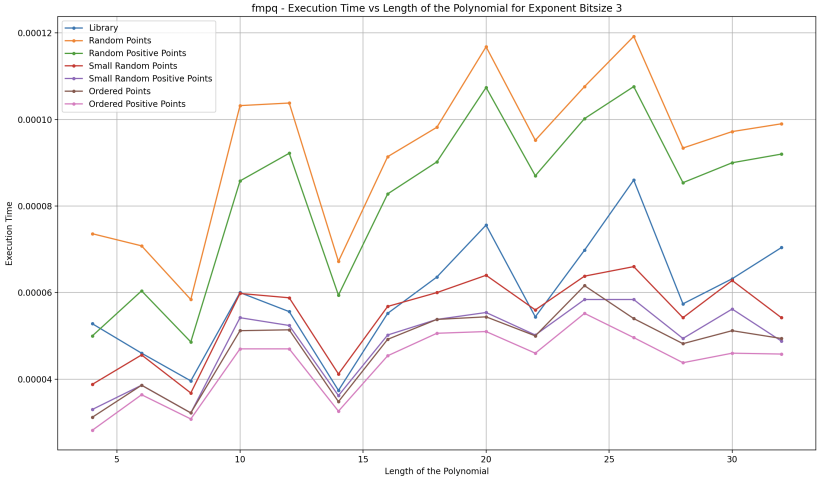
Difference becomes more important as the length of the polynomial & bitsize of the exponent increase.

FMPQ Benchmarks

Benchmark Setup

- Apple MacBook, 8-core Apple M2 SoC
- 16 GB unified memory
- Average of 5 independent trials
- Coefficient bit size fixed at 50
- Comparison against FLINT bivariate resultant computation

FMPQ Benchmarks



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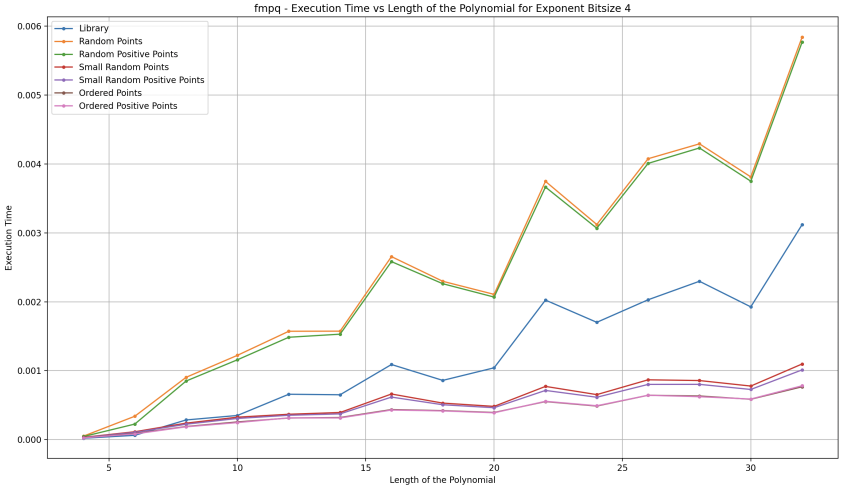

Bitsize 4

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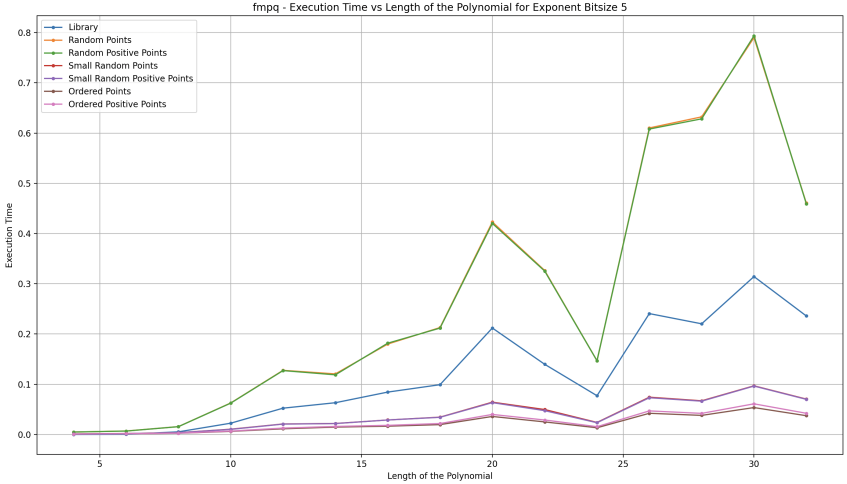
Execution time increases as the length of the polynomial increases.

FMPQ Benchmarks



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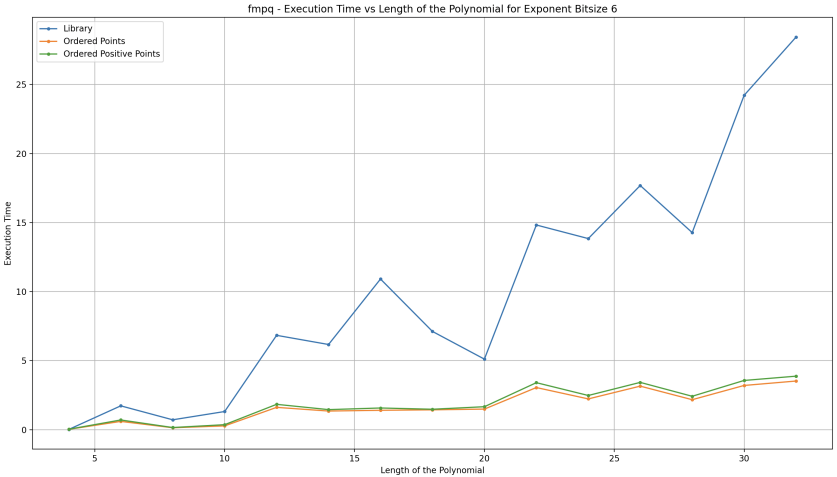

Bitsize 6

We remove random interpolation methods.

We observe 7x performance difference between our methods.

Behaviour is similar to $\mathbb{Z}$ mostly.

Conclusion

Summary of Results

- Subresultants control coefficient growth in GCD/resultant computations.
- Specialization is valid away from leading-coefficient zeros.
- Interpolation recovers multivariate resultants and subresultants.
- Evaluation point choice is critical for performance.
- Ordered or small-bit-size points work best in the benchmarks.
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Next Steps

- Extend implementation from resultants / subresultants to subresultant polynomials.
- Extend implementation to higher dimension multivariate polynomials (note that number of points for interpolation increases - $(D+1)^{n-1}$).
- Understand and implement specialization conditions for subresultant polynomials.
- Optimize, test, and benchmark for higher degree and length polynomials.